

Detecting autism using structural brain imaging data

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1 INTRODUCTION

Over the last few decades, diagnostic rates of autism spectrum disorder (ASD) have risen sharply. This rise is partly attributed to increased awareness of ASD. Nonetheless, the criteria used for diagnosis remain highly subjective and therefore depend strongly on the individual health care professional making the assessment. Consequently, researchers have sought to develop objective diagnostic tools based on brain imaging data, as the brains of autistic individuals exhibit structural differences. These data are then used to train various machine learning algorithms to perform diagnostic classification.

Despite these efforts, this strategy has achieved only limited success and reproducibility as proved by a 2015 literature review on the subject (Ha & al.). Indeed, different studies report widely varying accuracy levels, even when applying the same classification methods. This is due in part to the characteristics of the datasets that are used.

In this paper, we will employ structural brain data to construct a classifier with two primary goals: first, to classify patients with an accuracy comparable to that reported in a 2021 study by Wang, which uses the same dataset as we do; and second, to gain deeper insight into which specific brain features are most relevant for distinguishing between healthy control subjects and individuals diagnosed with autism.

2 DATA

To train our classifier, we relied on the ABIDE initiative dataset, which provides access to pre-processed brain imaging data on a freely available amazon bucket. Our dataset comprised more than 1035 subjects drawn from over 20+ acquisition sites, with site-level sample sizes ranging considerably, from as few as 10 subjects at the smallest contributing site to 60 at the largest. This imbalance is precisely what makes site harmonization so consequential for our results: a training fold that happens to draw only 1 or 2 subjects from an underrepresented site gives the model almost no chance to learn that site’s scanner-specific signal, which in turn hurts performance on that site’s subjects at test time. The subset we use is designed to be compatible with the Freesurfer imaging suite and includes measures of cortical thickness and gyration for each hemisphere, as well as volumetric information for each brain region derived from an atlas with 44 segmentation’s. For every subject, the data is divided into three separate files (for the right hemisphere, the left hemisphere and volumetric data). Each feature is accompanied by related statistical descriptors (mean, standard deviation, etc.).

In addition, we have phenotypic information for all participants. This includes diagnosis (indicating whether a participant is a healthy control or has ASD), age, sex, comorbidities, medications taken, verbal IQ, IQ, and more than 30 other characteristics per individual. Altogether, this amounts to over 840 data points for each patient.

Table 2.1: first 5 rows of our volumetric data for single patient

StructName	NumVert	SurfArea	GrayVol	ThickAvg	ThickStd	MeanCurv	GausCurv	FoldInd	CurvInd
bankssts	1559	1089	3081	2.732	0.533	0.124	0.033	18	2.0
caudalanteriorcingulate	1375	884	2420	2.417	0.705	0.130	0.032	21	1.8
caudalmiddlefrontal	4434	2660	7675	2.646	0.487	0.129	0.114	189	15.2
cuneus	2557	1621	3312	1.928	0.524	0.157	0.062	54	6.6
entorhinal	569	372	1888	3.495	0.852	0.126	0.055	9	1.2

Table 2.2: first 5 rows of our left hemispheric data for single patient

Index	SegId	NVoxels	Volume_mm3	StructName	normMean	normStdDev	normMin	normMax	normRange
1	4	17950	17950.0	Left-Lateral-Ventricle	36.7882	9.1888	20.0000	93.0000	73.0000
2	5	471	471.0	Left-Inf-Lat-Vent	51.7621	13.8728	24.0000	88.0000	64.0000
3	7	15498	15498.0	Left-Cerebellum-White-Matter	90.1014	4.8533	35.0000	103.0000	68.0000
4	8	55542	55542.0	Left-Cerebellum-Cortex	76.5028	9.0760	22.0000	103.0000	81.0000
5	10	7444	7444.0	Left-Thalamus-Proper	90.9855	8.1541	37.0000	110.0000	73.0000

3 METHOD

3.1 PREPROCESSING

Our goal is to start from three CSV files and end up with a structure that is suitable for training a neural network. More precisely, we construct a 2D NumPy array where each row contains all the information related to a single patient, including their diagnosis. We also chose not to include all the phenotypic data, as that would have made training much more costly, since columns like medication taken contain numerous drug entries. We would have had to use one-hot encoding to create columns for every medication in our dataset, which would have made our neural network much larger and more prone to saturation. As such, from the phenotypic data we only keep the age and sex of each patient, since our goal is to establish a diagnostic criterion based on the structural data of the brain.

Our data are then ready to be harmonized using neurocombat. Neurocombat is a Python library specifically designed to harmonize data across multiple sites. In our case, the MRI images on which the data are based were taken at more than 20 different sites. This harmonization step matters because MRI machines follow different specifications, and due to site-specific factors (MRI manufacturers, magnetic field strength, etc.), a model can end up learning to distinguish scanning sites rather than whether someone has autism. However, we leave out some key discrete variables, such as age and sex, from this harmonization step.

3.2 CLASSIFIER

Wang's paper describes a multi-head attention mechanism capable of inferring relationships between brain regions. The intuition behind this idea is that brain function for a given task is distributed across many areas, and if that function is impaired, we should see variations (declines) in brain volume and cortical thickness across all the areas involved in that function. However, attention transformers lack explainability (token-based methods are very hard to interpret) and are vastly more expensive in terms of resources and training time compared to a multilayer perceptron. Moreover, our tabular data is well suited for more classical techniques.

Our neural network is quite simple by design and necessity; it contains only two hidden layers and uses classic activation functions such as ReLU. Through trial and error, we noticed that using more layers decreased accuracy, which is a symptom of the vanishing gradient problem. We also normalize our outputs at every step to avoid saturating the network, using batch normalization. We chose to use the AdamW optimizer for its better regularization and generalization capabilities compared to the regular Adam optimizer which has a tendency to converge on local minimum. The default hyperparameter values yielded the best accuracy in our case and there was little need to use hyperparameter tuning techniques.

We used binary cross-entropy with a sigmoid activation on the final layer. That combo is pretty standard for two-class problems. The sigmoid squeezes the network's raw output into a probability between 0 and 1, so we can just read it directly. Then we set a threshold at 0.5, anything above that was classified autism (class 1), anything below will be a healthy control (class 0). Cross-entropy also has the useful property that it heavily penalizes the model when it's confidently wrong, which pushes the network to actually separate the two groups cleanly.

Because our model spits out probabilities, we can plug it into SHAP. The SHAP framework comes from game theory, it split the final prediction among all the input features. In practice, it takes each feature, changes its value, and watches how the model's output shifts. It does this across many different combinations of features, not just one at a time, to get a more accurate measure of each feature's real influence. Once it's done, we get a contribution value for every feature for every single patient.

We use this in two ways. First, we average the absolute contribution values across all subjects. That gives us a clean ranking of brain regions where we can see, for example, whether cortical thickness in the frontal lobe consistently matters more than gyrification in the temporal regions. Secondly, we can pull up one patient and see exactly which features pushed their probability up or down. Maybe for one individual, their left hemispheric volume pushed them toward autism, while for another it was a pattern of thinning in specific areas. This allows us to dig into the data and figure out which structural measurements discriminate between the groups (ASD patients vs healthy controls).

4 FINDINGS

Table 4.1: Confusion matrix

	Predicted TD	Predicted ASD
True TD	184	28
True ASD	38	164

Table 4.2: Classification report

	Precision	Recall	F1-score	Support
TD	0.83	0.87	0.85	212
ASD	0.85	0.81	0.83	202
Accuracy			0.84	414
Macro avg	0.84	0.84	0.84	414
Weighted avg	0.84	0.84	0.84	414

Table 4.3: Overall performance metrics

Metric	Value
Balanced accuracy	0.840
AUC-ROC	0.926
AUC-PR	0.907

The classification performance of the model was evaluated on a balanced set containing 414 samples (212 TD and 202 ASD). Overall, the model shows strong discriminative ability, reaching a total accuracy of 84%. The model correctly identified 184 TD individuals and 164 ASD individuals. Errors included 28 TD cases predicted as ASD (false positives) and 38 ASD cases misclassified as TD (false negatives). The higher number of false negatives suggests a slight tendency to classify ambiguous instances towards the TD group. It is however very difficult to determine the cause of our errors since we must also take into account the diagnostic errors made when labeling the data. If a doctor over-diagnoses a healthy test subject as autistic, we may still classify the patient correctly, but it would be recorded as a false negative. In this case, the skewed results indicate that we may have uncovered the presence of over-diagnosis. Our other metrics are also promising, for example with an AUC-ROC of 0.926 and an AUC-PR of 0.907 we know that we have a great discrimination ability. The close results between accuracy (0.84), balanced accuracy (0.840), and the macro/weighted average F1-scores (both 0.84) indicates that performance is stable and that we can most likely exclude over-fitting.

Table 4.4: Most important features

Rank	Feature	SHAP Values
1	rh_parsopercularis_GausCurv	0.1165
2	rh_rostralanteriorcingulate_CurvInd	0.0174
3	aseg_non-WM-hypointensities_Volume_mm3	0.0168
4	rh_inferiortemporal_GausCurv	0.0166
5	lh_lateralorbitofrontal_CurvInd	0.0145
6	rh_parahippocampal_CurvInd	0.0144
7	rh_paracentral_FoldInd	0.0142
8	lh_entorhinal_ThickStd	0.0130
9	rh_paracentral_ThickAvg	0.0122
10	lh_bankssts_MeanCurv	0.0117
11	lh_lateralorbitofrontal_FoldInd	0.0114
12	lh_cuneus_GausCurv	0.0113
13	lh_pericalcarine_FoldInd	0.0113
14	rh_caudalmiddlefrontal_ThickAvg	0.0111
15	aseg_CC_Central_normMax	0.0110
16	aseg_3rd-Ventricle_Volume_mm3	0.0110
17	lh_middletemporal_MeanCurv	0.0110
18	lh_posteriorcingulate_CurvInd	0.0110
19	aseg_CC_Mid_Anterior_normStdDev	0.0109
20	AGE_AT_SCAN	0.0108

Using SHAP, we also identified the right hemisphere’s pars opercularis as the single most predictive brain feature for autism in our model. More precisely, it is the Gaussian curvature of the pars opercularis. In regions with reduced gyrification, the corresponding gray matter volume is lower, since gray matter is predominantly located at the brain’s surface. This finding is consistent with a paper we found: a 2010 study reported reduced gray matter in the pars opercularis of autistic patients. Such a decrease in gray matter volume has been linked to diminished capacity for social cognition, inhibition, and prosody (how you say things), which helps explain several behavioral symptoms associated with autism. However, this feature’s SHAP value accounts for only about 12% of the combined contribution of our top 20 features, implying that the remaining 840 features should not be disregarded, as in some cases they may be equally or even more informative for diagnosis. In our program, it is often the case when we look at the SHAP analysis of single patients.

4.1 NEUROCOMBAT

As we’ve said before, neurocombat is used to correct site error. The main reason we use it is to remove site-specific noise. If we don’t, our accuracy drops to 68%, which makes sense

since we have data from over 20 different sites using different MRI machines, each with its own measurement quirks. If this isn't removed, our classifier ends up learning which scanner was used rather than whether or not the patient has ASD. Furthermore, our results change a lot when we train the model without neurocombat. This is mainly caused by the way our program samples data for training and testing, for example, if a training sample only has 1 or 2 subjects from a specific site, we can be almost certain that we'll get bad accuracy for that site when we evaluate on the test sample.

5 CONCLUSION

In this paper we set out to build a classifier that could tell autistic patients apart from healthy controls using structural brain data, with accuracy comparable to Wang's self-attention model. Our network reached a balanced accuracy of 84% and an AUC-ROC of 0.926, which shows that a much simpler architecture can do just as well without needing attention mechanisms.

Using SHAP also let us avoid the usual "black box" problem that comes with neural networks. We found that the Gaussian curvature of the right pars opercularis was the single most predictive feature, which lines up with earlier work linking this region to social cognition deficits in autism. But no single region carried the whole signal, autism seems to leave its mark across many different measures (thickness, curvature, volume) rather than in one specific spot. This fits with the idea that ASD is more about distributed brain connectivity than one isolated abnormality.

We also confirmed that harmonizing across our 20+ sites with neurocombat was very important to yield good results. Without it, our accuracy dropped from 84% to 68%. If this step is skipped you risk training a classifier that just learns to recognize scanners instead of autism, which is probably part of why the field has had so much trouble with reproducibility (Ha & al.).

That said, our model isn't perfect. Since we relied on the original phenotypic diagnoses as ground truth, any inconsistency in how ASD was diagnosed in the first place gets included into our results, this might explain part of our false negative rate. Going forward, it could be worth trying subgroup specific classifiers like Katuwal & al.

6 REFERENCES

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